Introduction

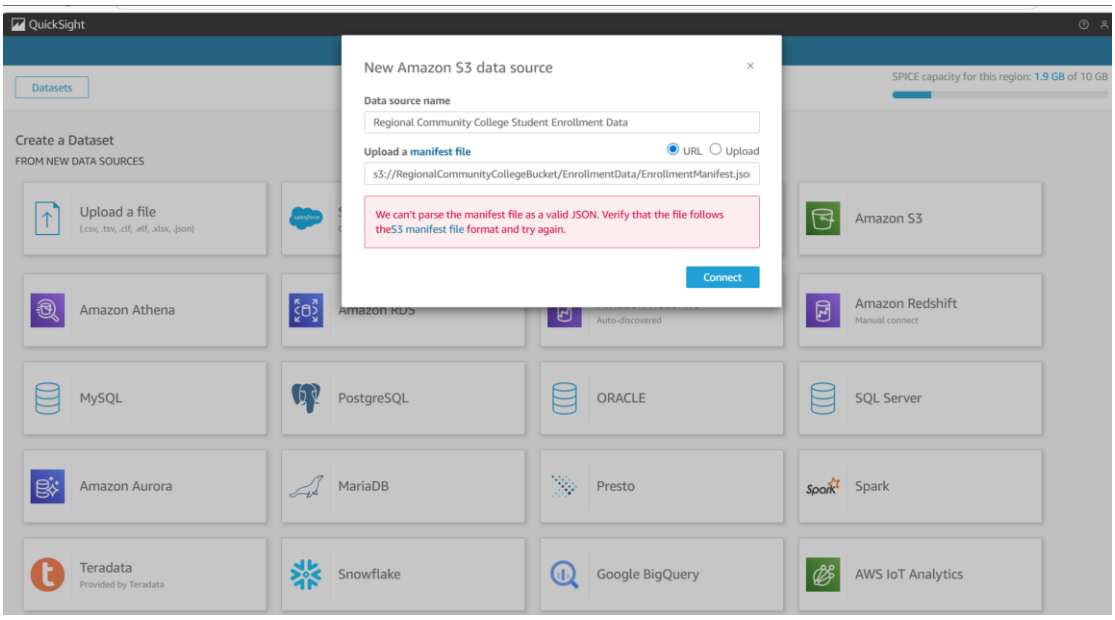
This project demonstrates how cloud analytics using AWS can help a college analyze student enrollment data and make better academic and financial decisions.
The main goal is to transform raw data into meaningful dashboards that support strategic planning.

Tools & Technologies

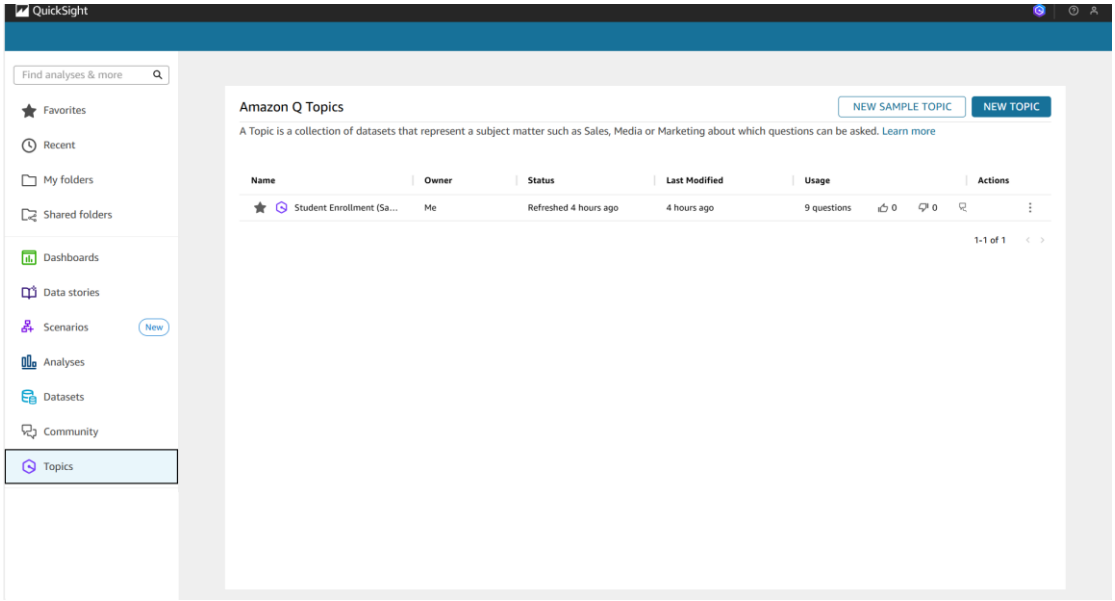
- Amazon S3 – Store raw student data
- Amazon Athena – Query data using SQL
- Amazon QuickSight – Create dashboards and visuals
- CSV Dataset – Student enrollment dataset

S3 Manifest Error

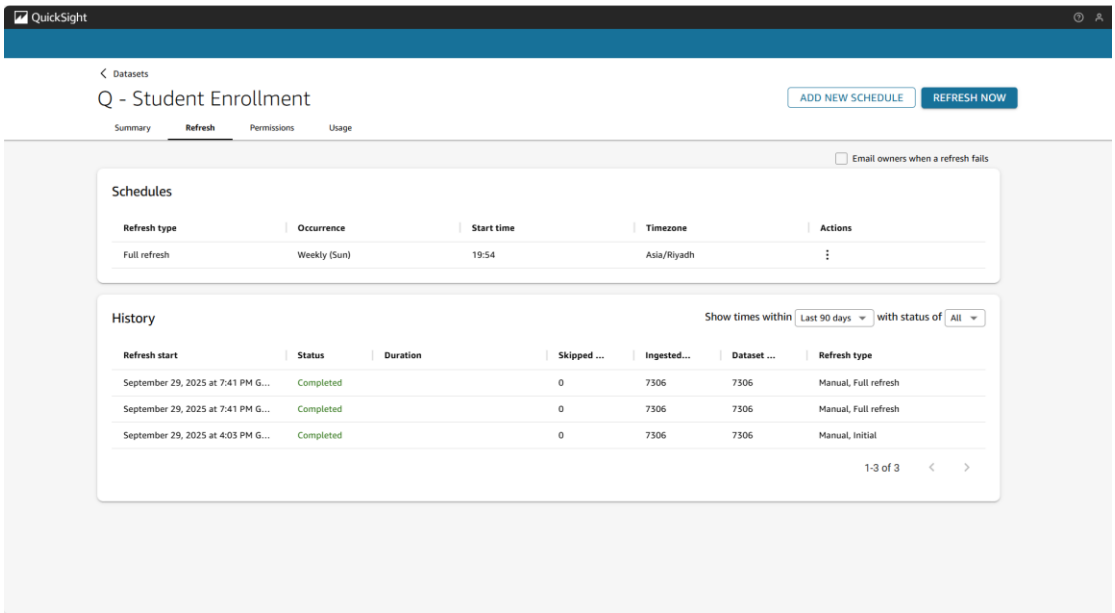
While uploading the dataset to S3, a manifest file error occurred.
This error was caused by an incorrect file path inside the manifest.
After fixing the path and updating the file, the dataset was successfully loaded.



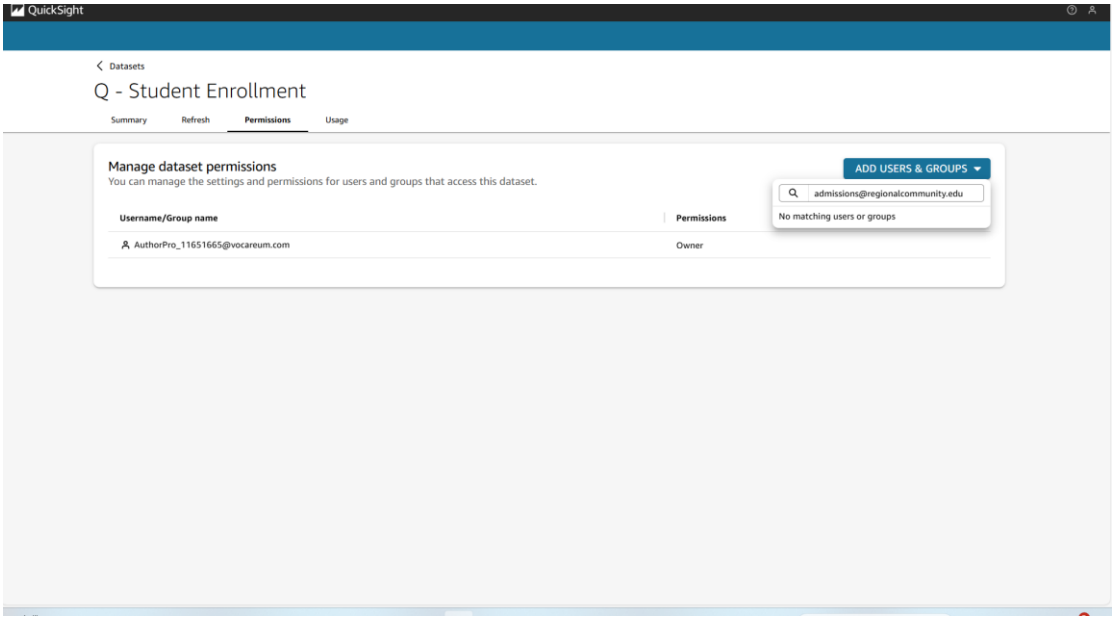
This screen shows the creation of the student enrollment table using the uploaded data

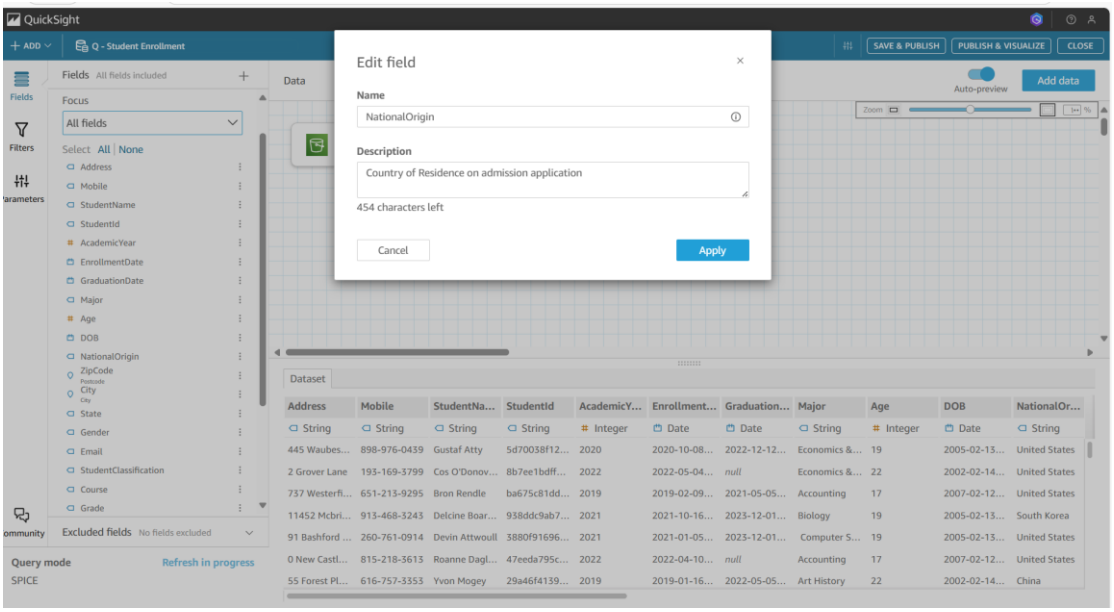


"03_Sample_Topic_Creation.png — Screen for creating the 'Student Enrollment Statistics' sample topic (in progress/completed)."



"Edit Email"


QuickSight
Datasets
Q - Student Enrollment
Summary
Refresh
Permissions
Usage
Manage dataset permissions
You can manage the settings and permissions for users and groups that access this dataset.
ADD USERS & GROUPS
admissions@regionalcommunity.edu
No matching users or groups
Username/Group name
Permissions
AuthorPro_11651665@vocareum.com
Owner



"Here, the name was changed and modified."

Add calculated field

Student Type

Cancel Save

```
1 ifelse({Age} < 30, 'Youth', 'Adult Continuing Education')
2
```

Fields

Parameters

Functions

Search functions

All

abs
addDateTime
addWorkDays
avg
avgIf
avgOver
ceil
coalesce

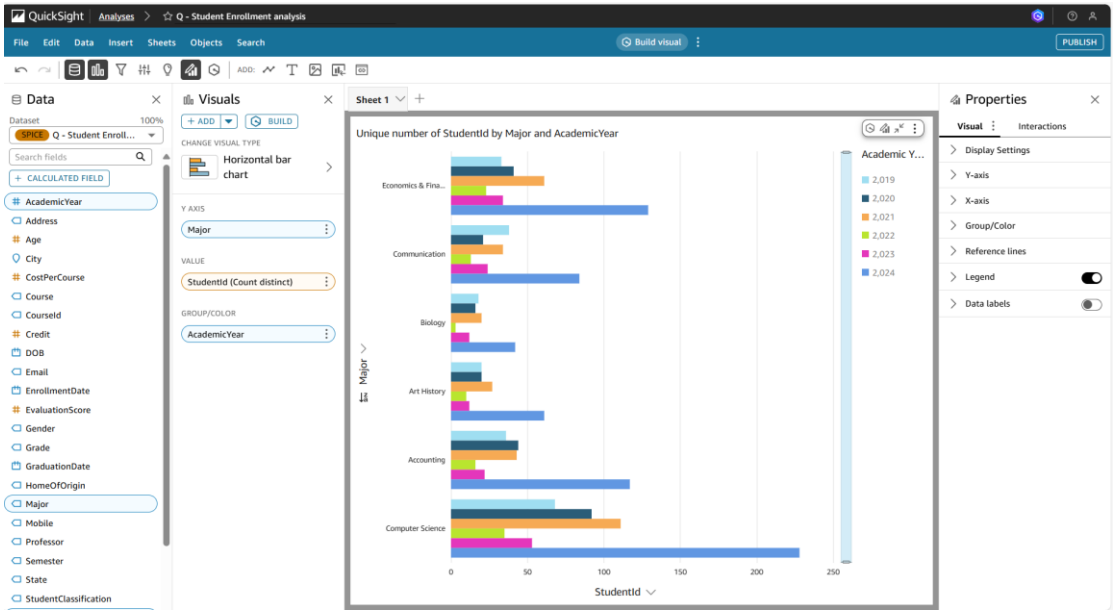
abs

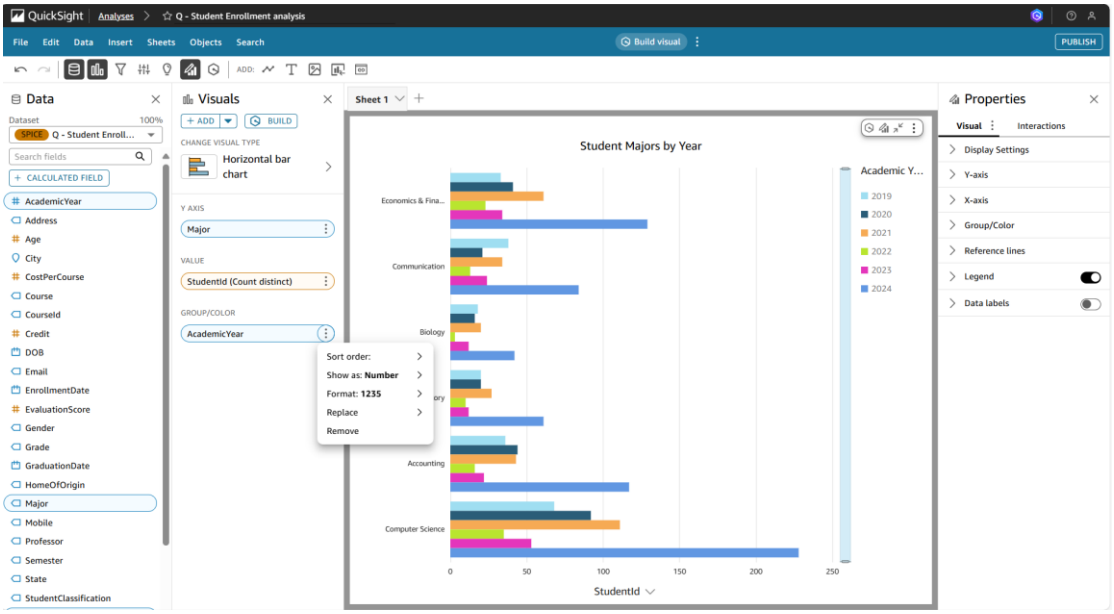
Returns the absolute value of a given expression.

SYNTAX

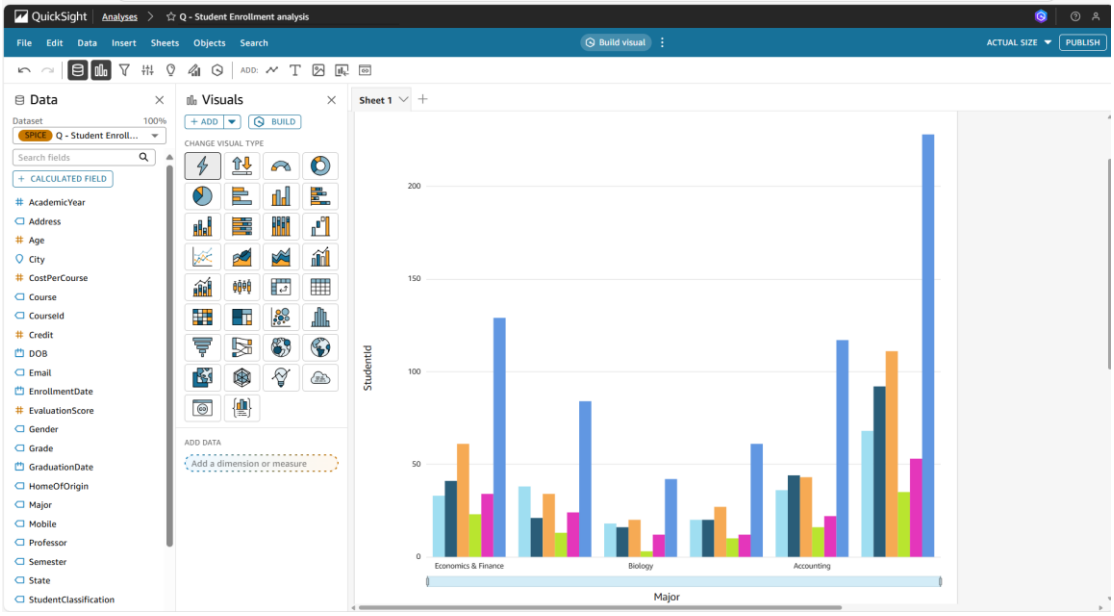
abs(*expression*)

Learn More

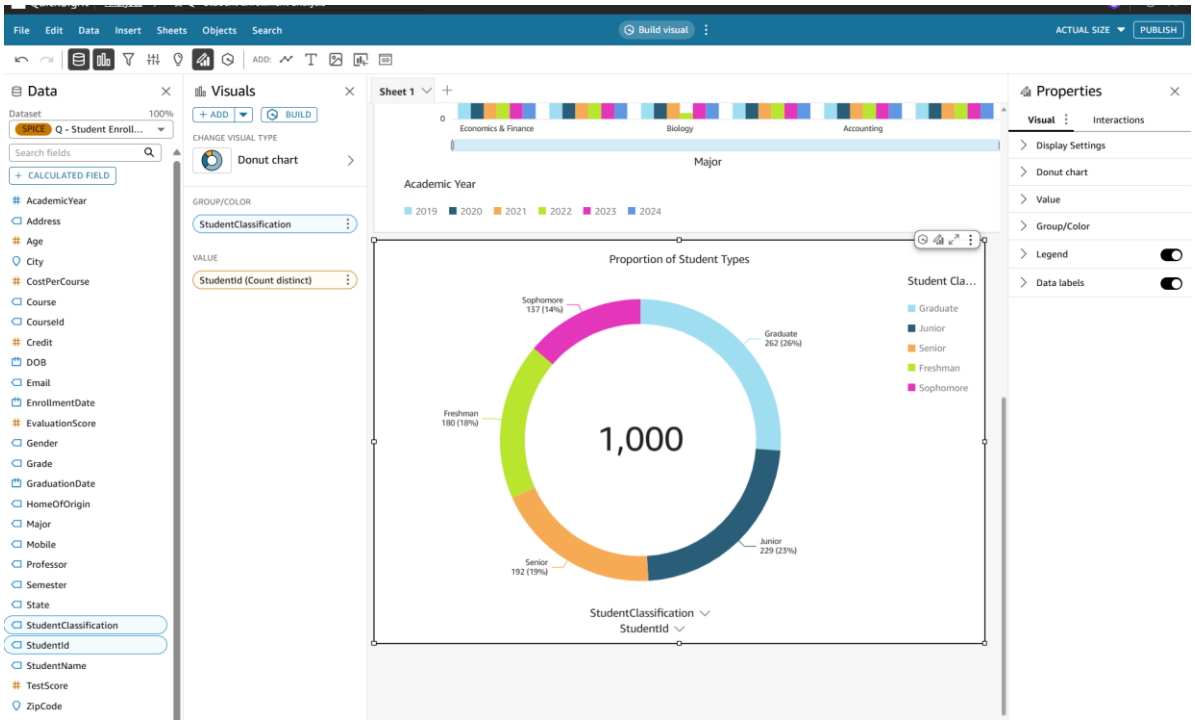

QuickSight
Analyses
Q - Student Enrollment analysis
File
Edit
Data
Insert
Sheets
Objects
Search
Build visual
PUBLISH
Data
Dataset
100%
SPICE Q - Student Enroll...
Search fields
+ CALCULATED FIELD
AcademicYear
Address
Age
City
CostPerCourse
Course
CourseId
Credit
DOB
Email
EnrollmentDate
EvaluationScore
Gender
Grade
GraduationDate
HomeOfOrigin
Major
Mobile
Professor
Semester
State
StudentClassification
Visuals
+ ADD
BUILD
CHANGE VISUAL TYPE
Horizontal bar chart
Y AXIS
Major
VALUE
StudentId (Count distinct)
GROUP/COLOR
AcademicYear
Sheet 1
Unique number of StudentId by Major and AcademicYear
Academic Y...
2,019
2,020
2,021
2,022
2,023
2,024
Economics & Fina...
Communication
Biology
Art History
Accounting
Computer Science
Major
0
50
100
150
200
250
StudentId
Properties
Visual
Interactions
Display Settings
Y-axis
X-axis
Group/Color
Reference lines
Legend
Data labels

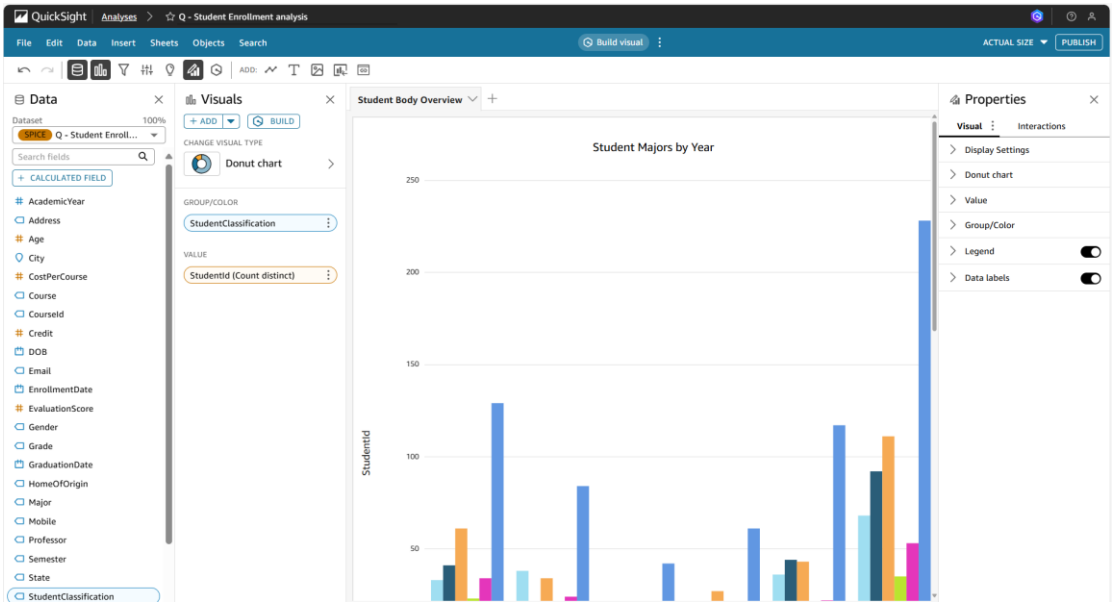

QuickSight
Analyses
Q - Student Enrollment analysis
File
Edit
Data
Insert
Sheets
Objects
Search
Build visual
PUBLISH
Data
Dataset
100%
SPICE
Q - Student Enroll...
Search fields
CALCULATED FIELD
AcademicYear
Address
Age
City
CostPerCourse
Course
CourseId
Credit
DOB
Email
EnrollmentDate
EvaluationScore
Gender
Grade
GraduationDate
HomeOfOrigin
Major
Mobile
Professor
Semester
State
StudentClassification
Visuals
ADD
BUILD
CHANGE VISUAL TYPE
Horizontal bar chart
Y AXIS
Major
VALUE
StudentId (Count distinct)
GROUP/COLOR
AcademicYear
Sort order:
Show as: Number
Format: 1235
Replace
Remove
Sheet 1
Student Majors by Year
Economics & Fina...
Communication
Biology
Accounting
Computer Science
0
50
100
150
200
250
StudentId
Academic Y...
2019
2020
2021
2022
2023
2024
Properties
Visual
Interactions
Display Settings
Y-axis
X-axis
Group/Color
Reference lines
Legend
Data labels

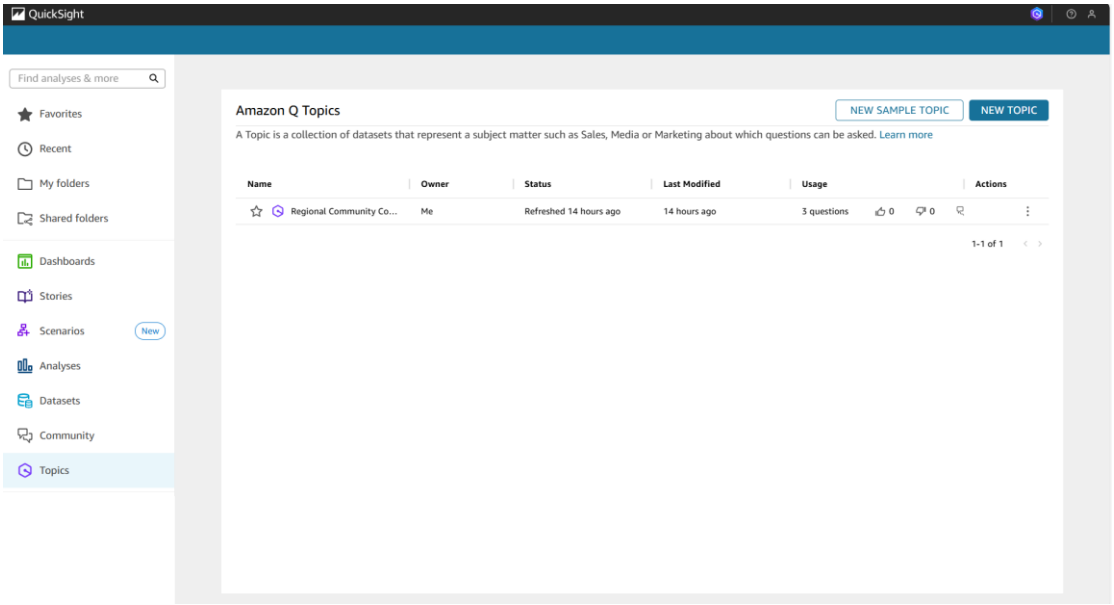
Add the visual chart and modify it

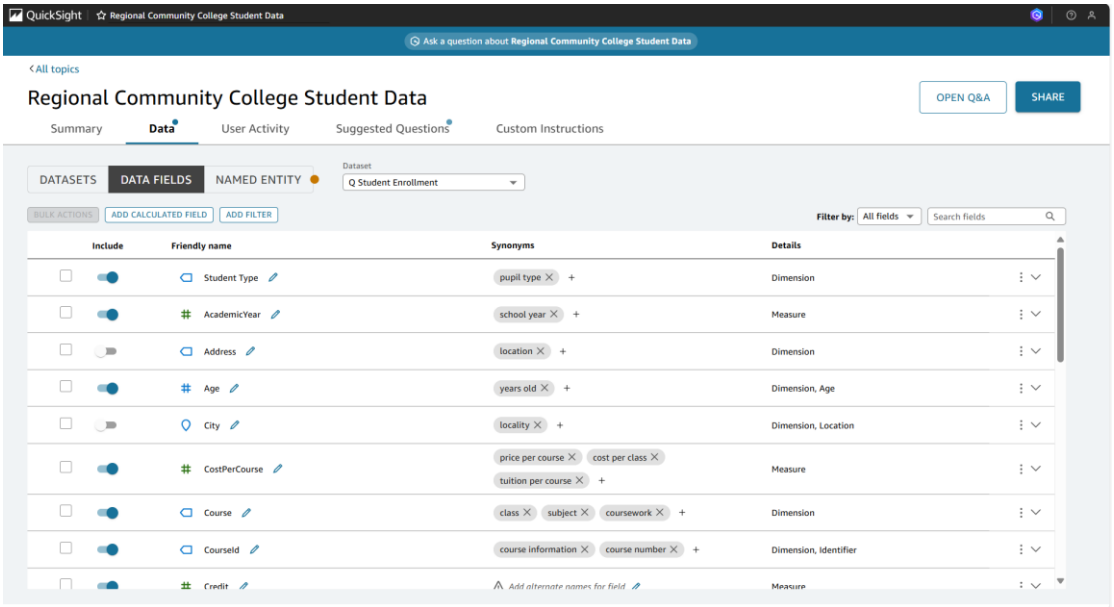


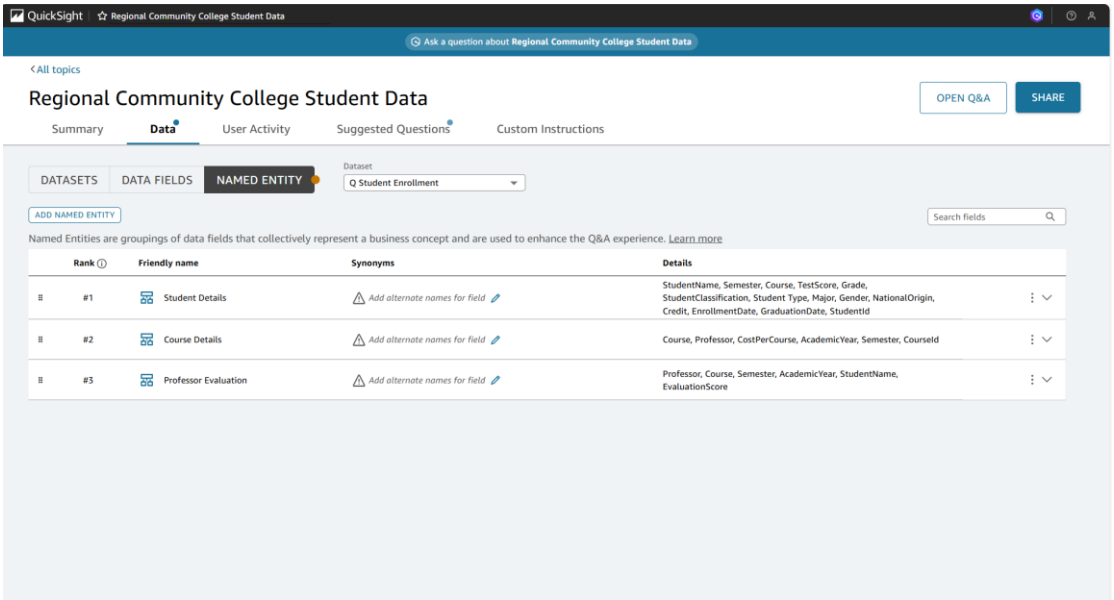
"And this is the second visual"

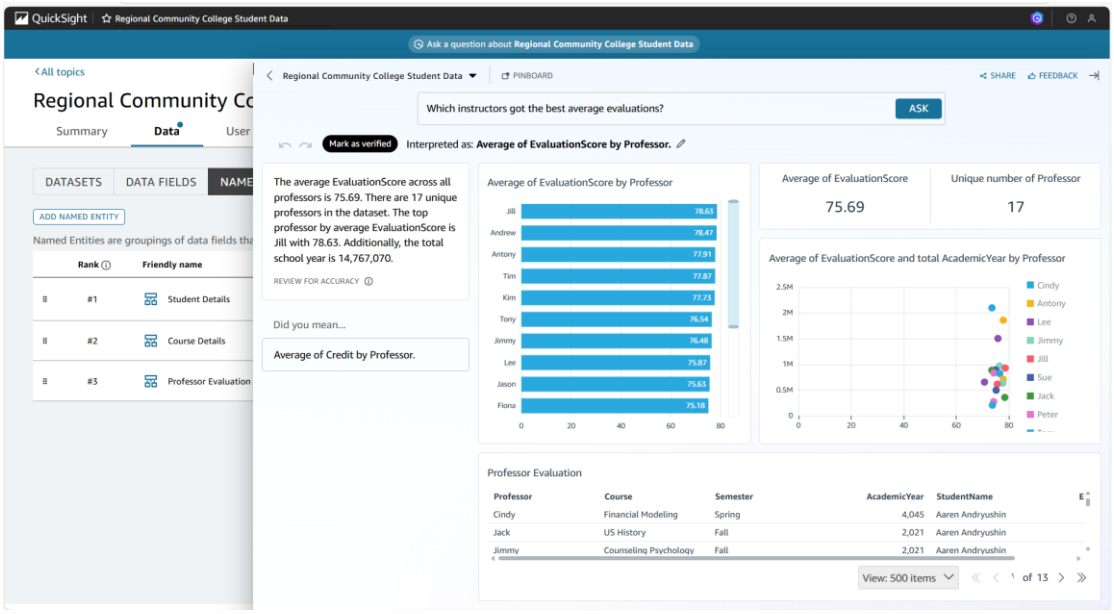


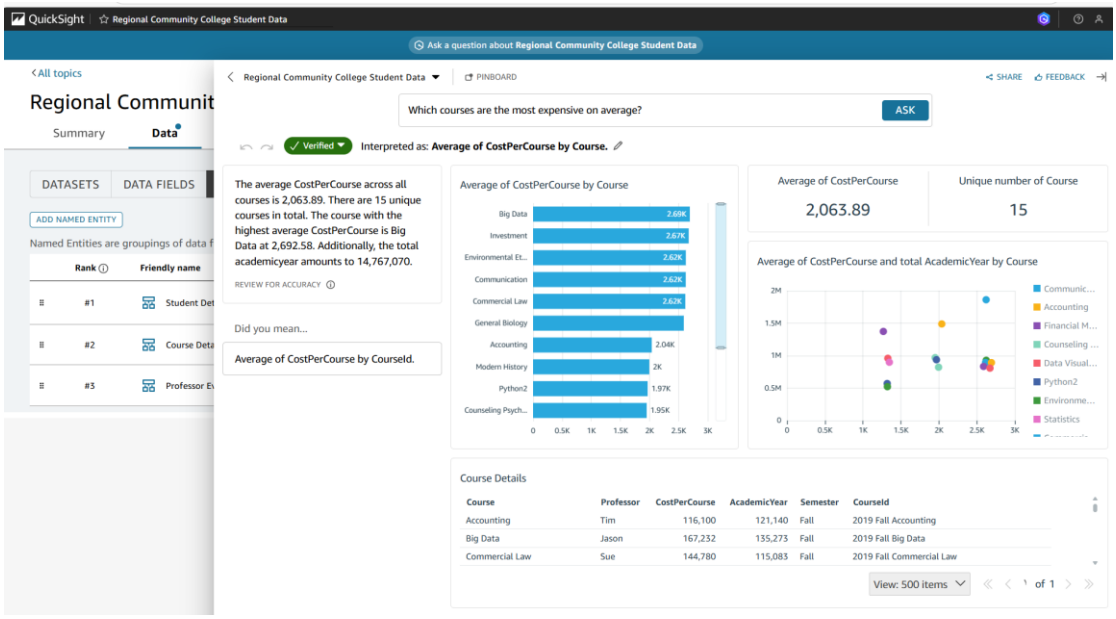

QuickSight
Analyses
Q - Student Enrollment analysis
File
Edit
Data
Insert
Sheets
Objects
Search
Build visual
ACTUAL SIZE
PUBLISH
Data
Dataset
100%
SPICE
Q - Student Enroll...
Search fields
CALCULATED FIELD
AcademicYear
Address
Age
City
CostPerCourse
Course
CourseId
Credit
DOB
Email
EnrollmentDate
EvaluationScore
Gender
Grade
GraduationDate
HomeOfOrigin
Major
Mobile
Professor
Semester
State
StudentClassification
Visuals
ADD
BUILD
CHANGE VISUAL TYPE
Donut chart
GROUP/COLOR
StudentClassification
VALUE
StudentId (Count distinct)
Student Body Overview
Student Majors by Year
StudentId
250
200
150
100
50
Properties
Visual
Interactions
Display Settings
Donut chart
Value
Group/Color
Legend
Data labels


QuickSight
Find analyses & more
Favorites
Recent
My folders
Shared folders
Dashboards
Stories
Scenarios
New
Analyses
Datasets
Community
Topics
Amazon Q Topics
NEW SAMPLE TOPIC
NEW TOPIC
A Topic is a collection of datasets that represent a subject matter such as Sales, Media or Marketing about which questions can be asked. Learn more
Name
Owner
Status
Last Modified
Usage
Actions
Regional Community Co...
Me
Refreshed 14 hours ago
14 hours ago
3 questions
0
0
1-1 of 1

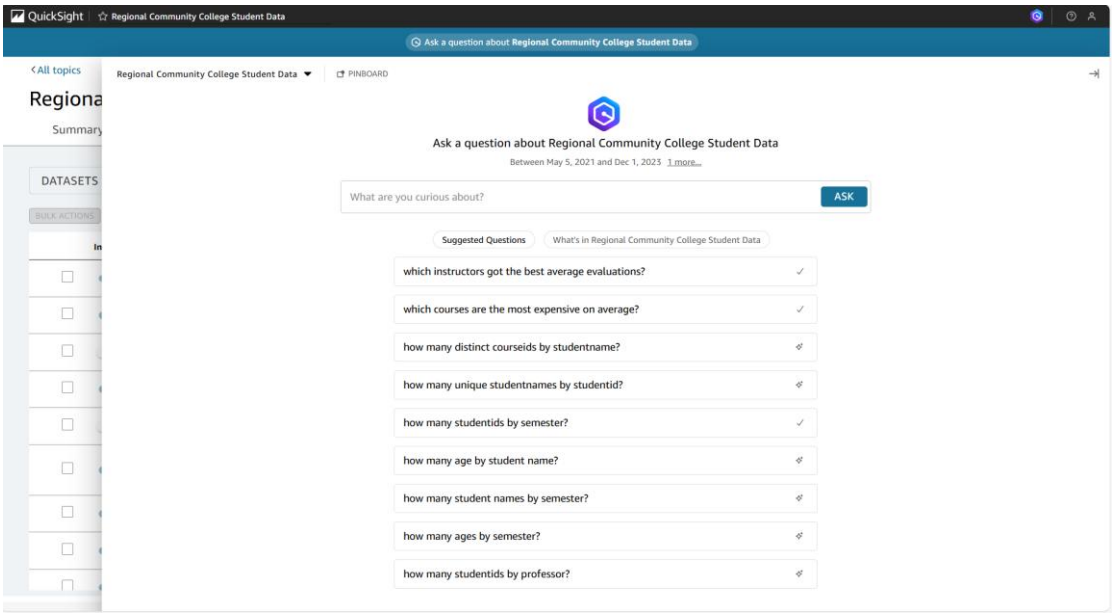

QuickSight
Regional Community College Student Data
Ask a question about Regional Community College Student Data
All topics
Regional Community College Student Data
OPEN Q&A
SHARE
Summary
Data
User Activity
Suggested Questions
Custom Instructions
DATASETS
DATA FIELDS
NAMED ENTITY
Dataset
Q Student Enrollment
BULK ACTIONS
ADD CALCULATED FIELD
ADD FILTER
Filter by:
All fields
Search fields
Include
Friendly name
Synonyms
Details
Student Type
pupil type
Dimension
AcademicYear
school year
Measure
Address
location
Dimension
Age
years old
Dimension, Age
City
locality
Dimension, Location
CostPerCourse
price per course
cost per class
tuition per course
Measure
Course
class
subject
coursework
Dimension
CourseId
course information
course number
Dimension, Identifier
Credit
Add alternate names for field
Measure

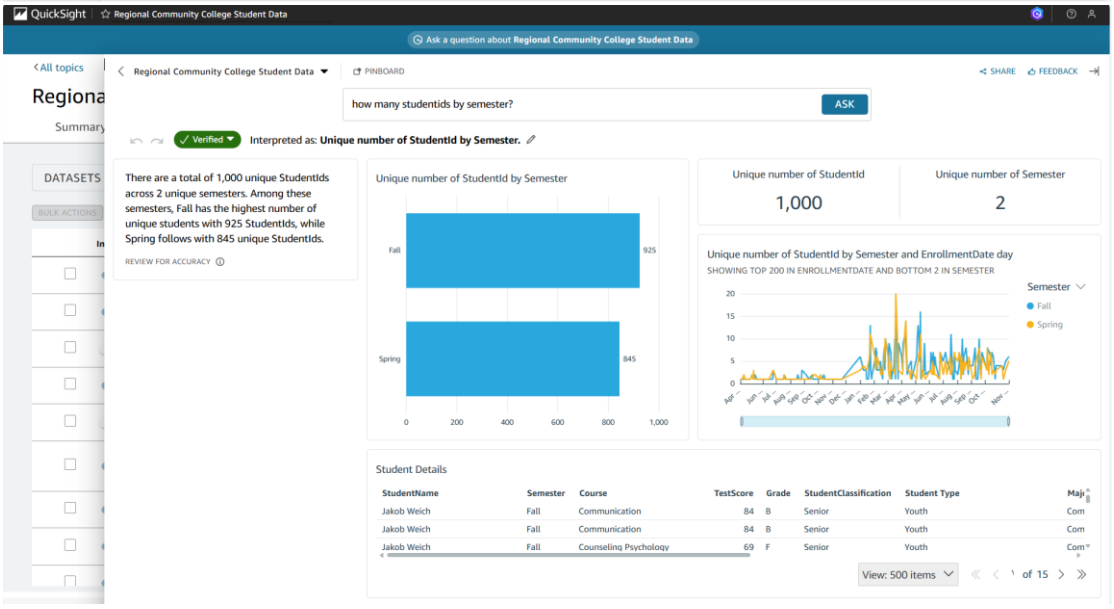

QuickSight
Regional Community College Student Data
Ask a question about Regional Community College Student Data
All topics
Regional Community College Student Data
OPEN Q&A
SHARE
Summary
Data
User Activity
Suggested Questions
Custom Instructions
DATASETS
DATA FIELDS
NAMED ENTITY
Dataset
Q Student Enrollment
ADD NAMED ENTITY
Search fields
Named Entities are groupings of data fields that collectively represent a business concept and are used to enhance the Q&A experience. Learn more
Rank
Friendly name
Synonyms
Details
#1
Student Details
Add alternate names for field
StudentName, Semester, Course, TestScore, Grade, StudentClassification, Student Type, Major, Gender, NationalOrigin, Credit, EnrollmentDate, GraduationDate, StudentId
#2
Course Details
Add alternate names for field
Course, Professor, CostPerCourse, AcademicYear, Semester, CourseId
#3
Professor Evaluation
Add alternate names for field
Professor, Course, Semester, AcademicYear, StudentName, EvaluationScore

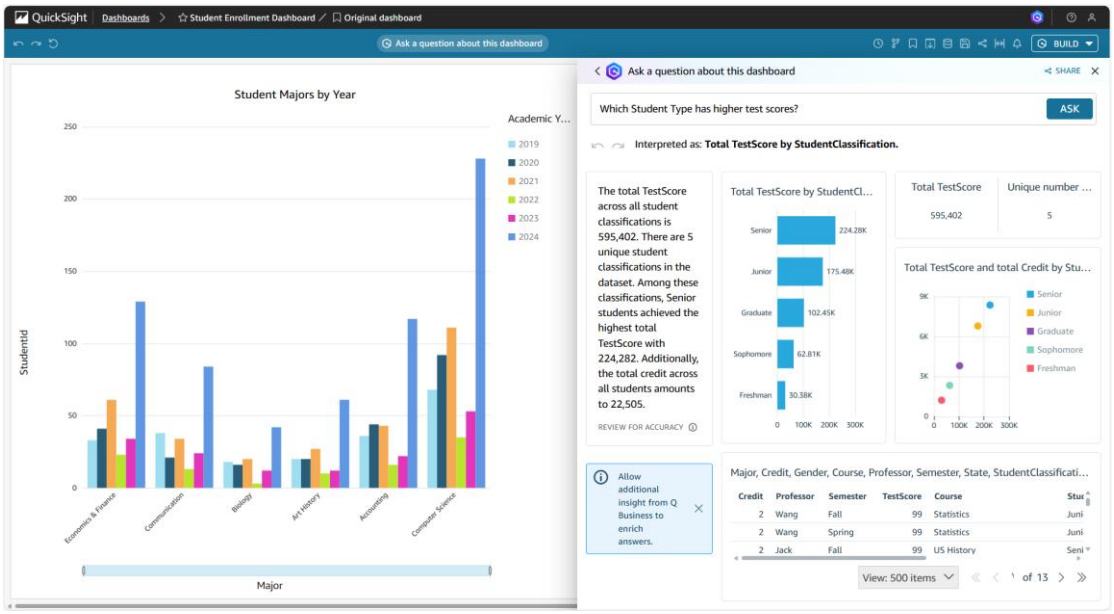

QuickSight
Regional Community College Student Data
Ask a question about Regional Community College Student Data
All topics
Regional Community Co
Summary
Data
User
DATASETS
DATA FIELDS
NAME
ADD NAMED ENTITY
Named Entities are groupings of data fields tha
Rank
Friendly name
#1
Student Details
#2
Course Details
#3
Professor Evaluation
Regional Community College Student Data
PINBOARD
SHARE
FEEDBACK
Which instructors got the best average evaluations?
ASK
Mark as verified
Interpreted as: Average of EvaluationScore by Professor.
The average EvaluationScore across all professors is 75.69. There are 17 unique professors in the dataset. The top professor by average EvaluationScore is Jill with 78.63. Additionally, the total school year is 14,767,070.
REVIEW FOR ACCURACY
Did you mean...
Average of Credit by Professor.
Average of EvaluationScore by Professor
Jill 78.63
Andrew 78.47
Antony 77.91
Tim 77.87
Kim 77.73
Tony 76.54
Jimmy 76.48
Lee 75.87
Jason 75.65
Fiona 75.18
0 20 40 60 80
Average of EvaluationScore
75.69
Unique number of Professor
17
Average of EvaluationScore and total AcademicYear by Professor
2.5M
2M
1.5M
1M
0.5M
0
0 20 40 60 80
Cindy
Antony
Lee
Jimmy
Jill
Sue
Jack
Peter
Professor Evaluation
Professor Course Semester AcademicYear StudentName
Cindy Financial Modeling Spring 4,045 Aaren Andryushin
Jack US History Fall 2,021 Aaren Andryushin
Jimmy Counseling Psychology Fall 2,021 Aaren Andryushin
View: 500 items
of 13

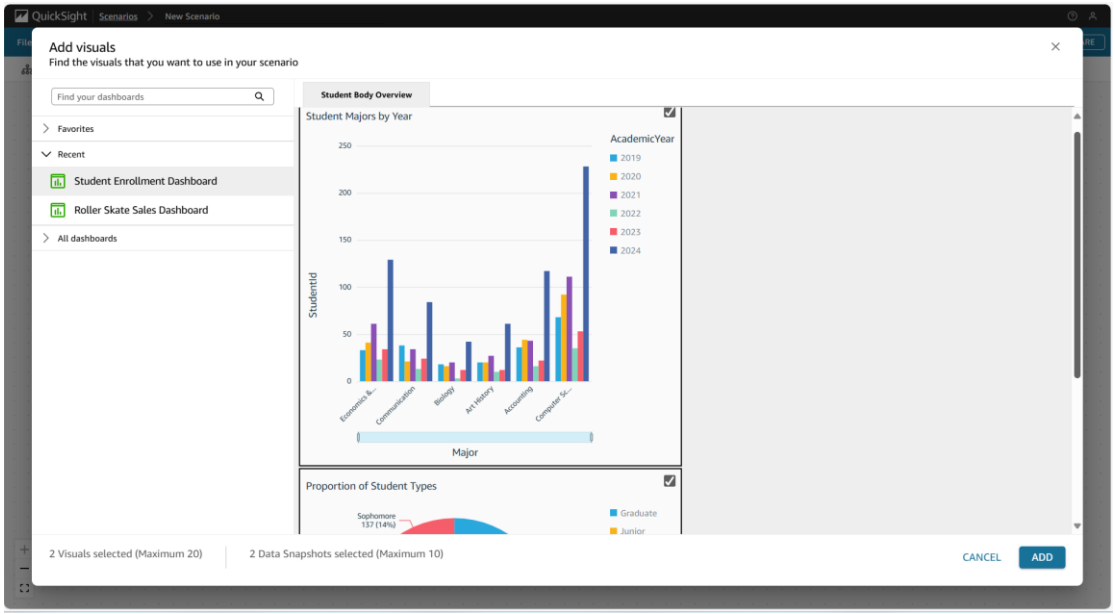


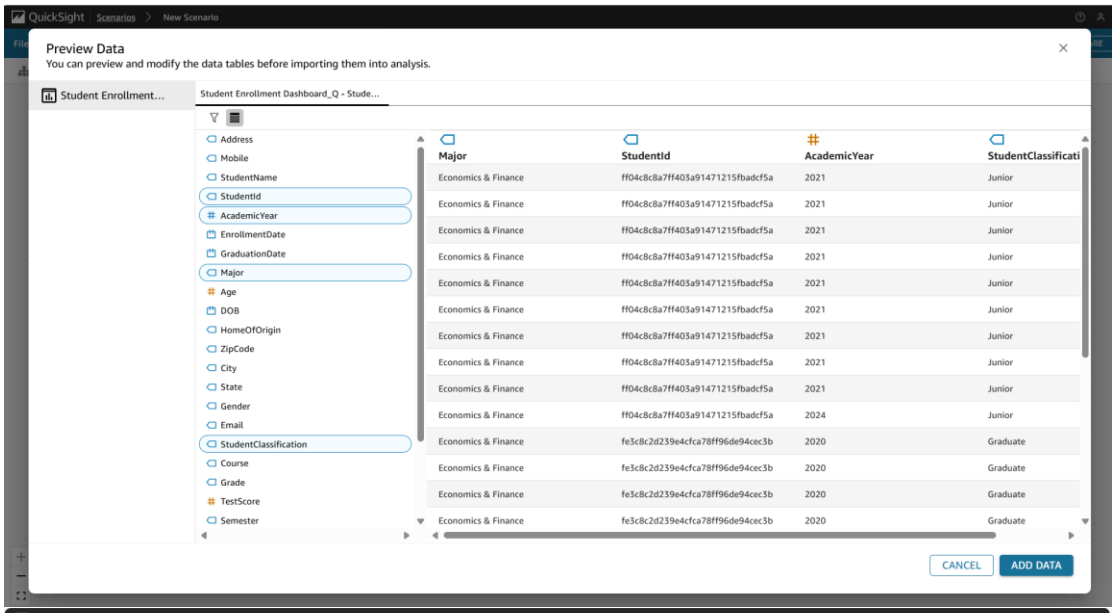
Course	Professor	CostPerCourse	AcademicYear	Semester	CourseId
Accounting	Tim	116,100	121,140	Fall	2019 Fall Accounting
Big Data	Jason	167,232	135,273	Fall	2019 Fall Big Data
Commercial Law	Sue	144,780	115,083	Fall	2019 Fall Commercial Law

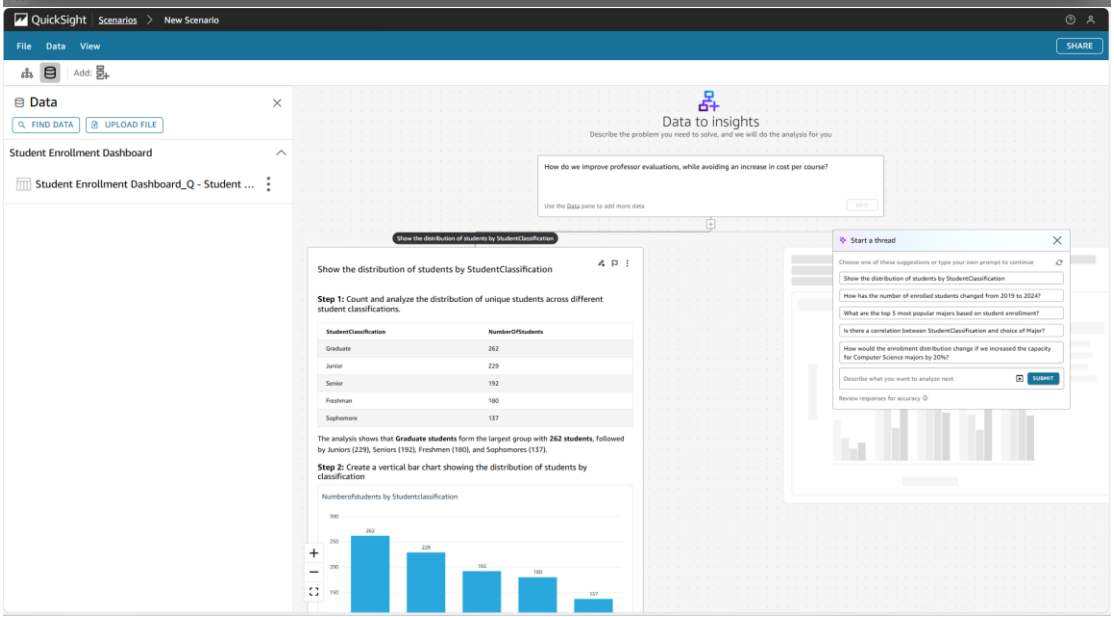

QuickSight
Regional Community College Student Data
Ask a question about Regional Community College Student Data
All topics
Regiona
Summary
DATASETS
BULK ACTIONS
Regional Community College Student Data
PINBOARD
Ask a question about Regional Community College Student Data
Between May 5, 2021 and Dec 1, 2023 1 more...
What are you curious about?
ASK
Suggested Questions
What's in Regional Community College Student Data
which instructors got the best average evaluations?
which courses are the most expensive on average?
how many distinct courseids by studentname?
how many unique studentnames by studentid?
how many studentids by semester?
how many age by student name?
how many student names by semester?
how many ages by semester?
how many studentids by professor?

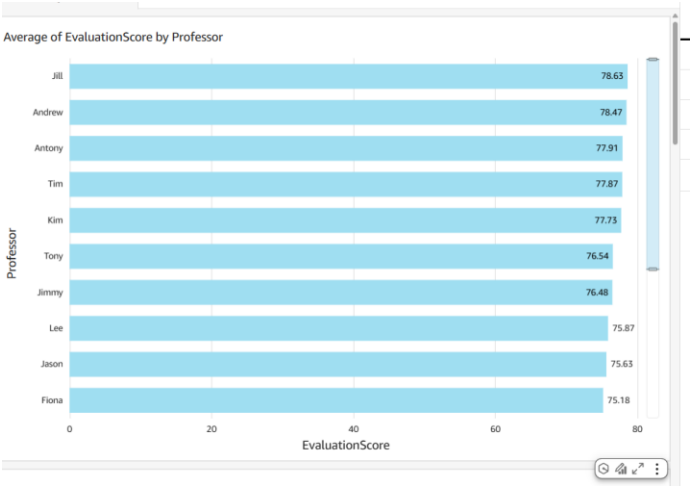

QuickSight
Regional Community College Student Data
Ask a question about Regional Community College Student Data
All topics
Regiona
Summary
DATASETS
BULK ACTIONS
Regional Community College Student Data
PINBOARD
SHARE
FEEDBACK
how many studentids by semester?
ASK
Verified
Interpreted as: Unique number of StudentId by Semester.
There are a total of 1,000 unique StudentIds across 2 unique semesters. Among these semesters, Fall has the highest number of unique students with 925 StudentIds, while Spring follows with 845 unique StudentIds.
REVIEW FOR ACCURACY
Unique number of StudentId by Semester
Fall
925
Spring
845
0
200
400
600
800
1,000
Unique number of StudentId
1,000
Unique number of Semester
2
Unique number of StudentId by Semester and EnrollmentDate day
SHOWING TOP 200 IN ENROLLMENTDATE AND BOTTOM 2 IN SEMESTER
Semester
Fall
Spring
Student Details
StudentName
Semester
Course
TestScore
Grade
StudentClassification
Student Type
Jakob Weich
Fall
Communication
84
B
Senior
Youth
Jakob Weich
Fall
Communication
84
B
Senior
Youth
Jakob Weich
Fall
Counseling Psychology
69
F
Senior
Youth
View: 500 items
of 15

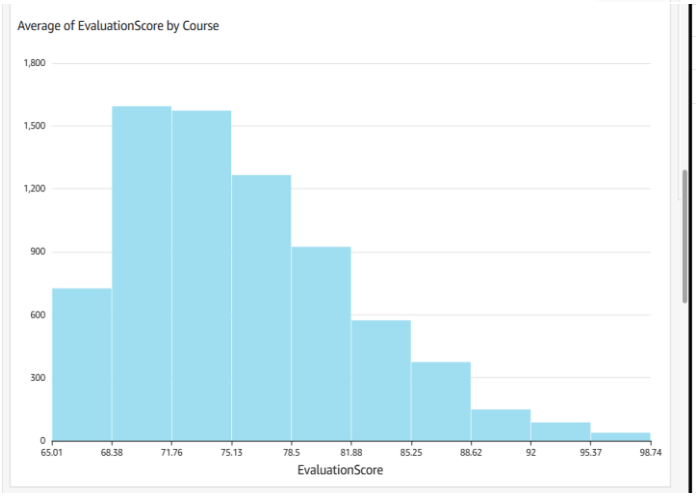

QuickSight
Dashboards
Student Enrollment Dashboard
Original dashboard
Ask a question about this dashboard
BUILD
Student Majors by Year
Academic Y...
2019
2020
2021
2022
2023
2024
Studentid
Economics & Finance
Communication
Biology
Art History
Accounting
Computer Science
Major
Ask a question about this dashboard
SHARE
Which Student Type has higher test scores?
ASK
Interpreted as: Total TestScore by StudentClassification.
The total TestScore across all student classifications is 595,402. There are 5 unique student classifications in the dataset. Among these classifications, Senior students achieved the highest total TestScore with 224,282. Additionally, the total credit across all students amounts to 22,505.
REVIEW FOR ACCURACY
Total TestScore by StudentCl...
Senior
224.28K
Junior
175.48K
Graduate
102.45K
Sophomore
62.81K
Freshman
30.38K
Total TestScore
595,402
Unique number ...
5
Total TestScore and total Credit by Stu...
Senior
Junior
Graduate
Sophomore
Freshman
Allow additional insight from Q Business to enrich answers.
Major, Credit, Gender, Course, Professor, Semester, State, StudentClassificati...
Credit Professor Semester TestScore Course
2 Wang Fall 99 Statistics
2 Wang Spring 99 Statistics
2 Jack Fall 99 US History
View: 500 items
of 13


QuickSight
Scenarios
New Scenario
Add visuals
Find the visuals that you want to use in your scenario
Find your dashboards
Favorites
Recent
Student Enrollment Dashboard
Roller Skate Sales Dashboard
All dashboards
Student Body Overview
Student Majors by Year
AcademicYear
2019
2020
2021
2022
2023
2024
StudentId
Major
Proportion of Student Types
Graduate
Junior
2 Visuals selected (Maximum 20)
2 Data Snapshots selected (Maximum 10)
CANCEL
ADD

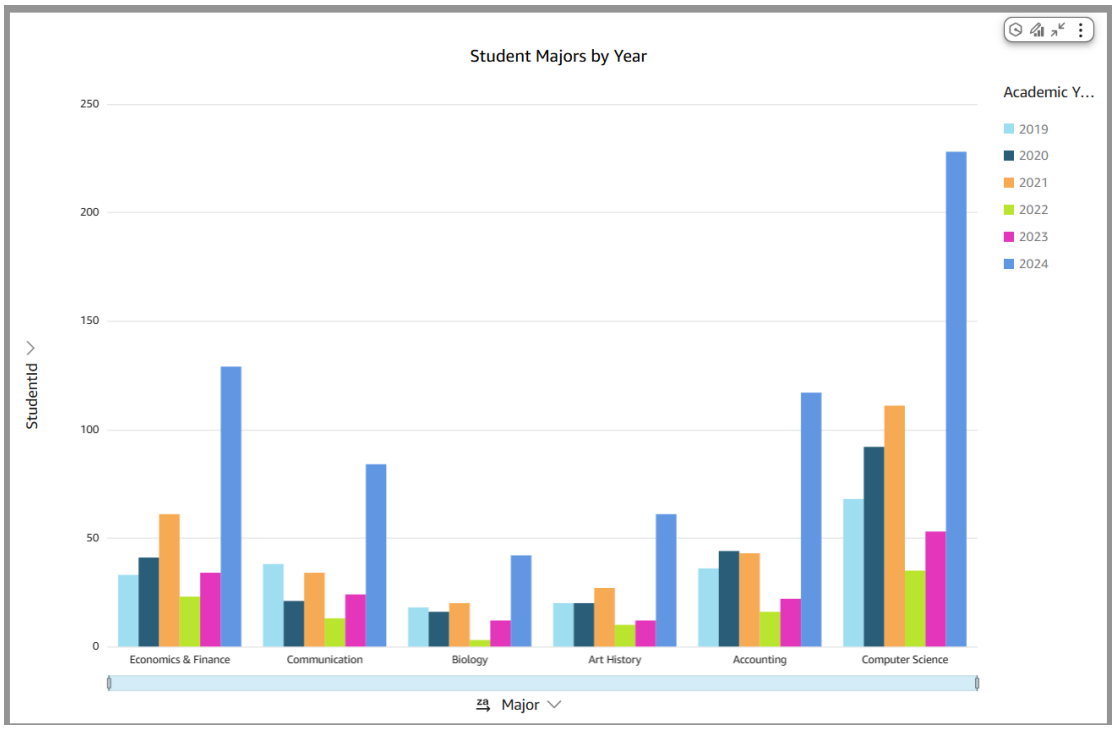

QuickSight
Scenarios
New Scenario
Preview Data
You can preview and modify the data tables before importing them into analysis.
Student Enrollment...
Student Enrollment Dashboard_Q - Stude...
Address
Mobile
StudentName
StudentId
AcademicYear
EnrollmentDate
GraduationDate
Major
Age
DOB
HomeOfOrigin
ZipCode
City
State
Gender
Email
StudentClassification
Course
Grade
TestScore
Semester
Major
StudentId
AcademicYear
StudentClassificati
CANCEL
ADD DATA


QuickSight
Scenarios
New Scenario
File
Data
View
SHARE
Data
FIND DATA
UPLOAD FILE
Student Enrollment Dashboard
Student Enrollment Dashboard_Q - Student ...
Data to insights
Describe the problem you need to solve, and we will do the analysis for you
How do we improve professor evaluations, while avoiding an increase in cost per course?
Show the distribution of students by StudentClassification
Start a thread
Show the distribution of students by StudentClassification
Step 1: Count and analyze the distribution of unique students across different student classifications.
Step 2: Create a vertical bar chart showing the distribution of students by classification

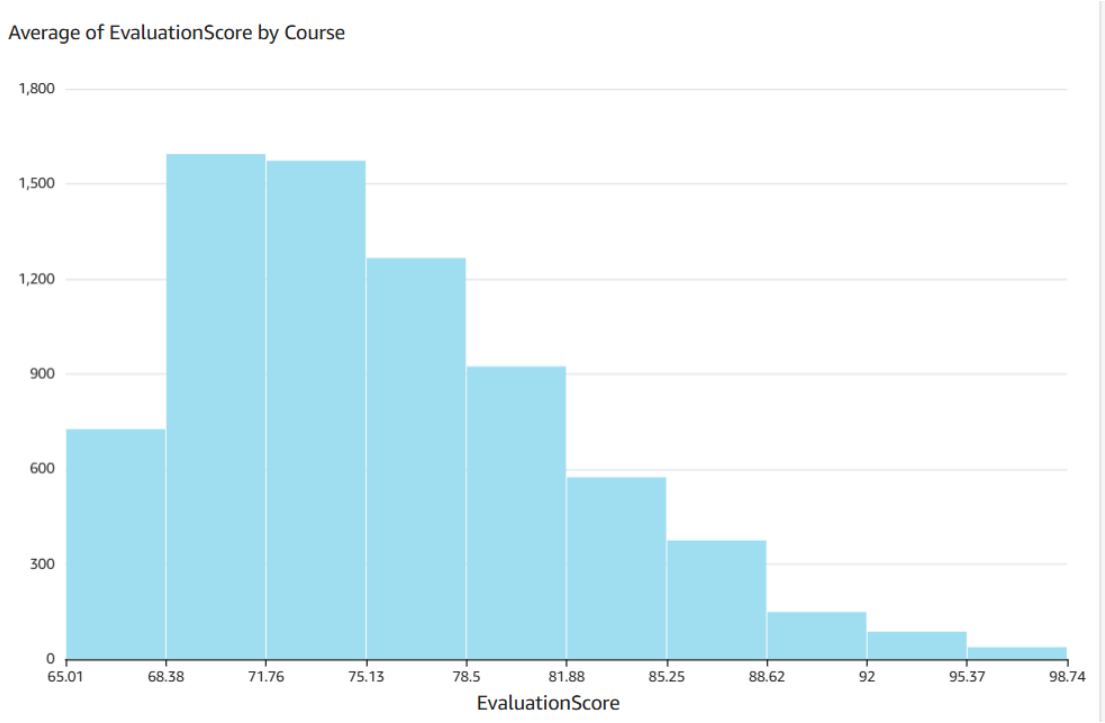

Average of EvaluationScore by Professor
Professor
Jill
Andrew
Antony
Tim
Kim
Tony
Jimmy
Lee
Jason
Fiona
78.63
78.47
77.91
77.87
77.73
76.54
76.48
75.87
75.63
75.18
0
20
40
60
80
EvaluationScore


Average of EvaluationScore by Course
1,800
1,500
1,200
900
600
300
0
65.01
68.38
71.76
75.13
78.5
81.88
85.25
88.62
92
95.37
98.74
EvaluationScore

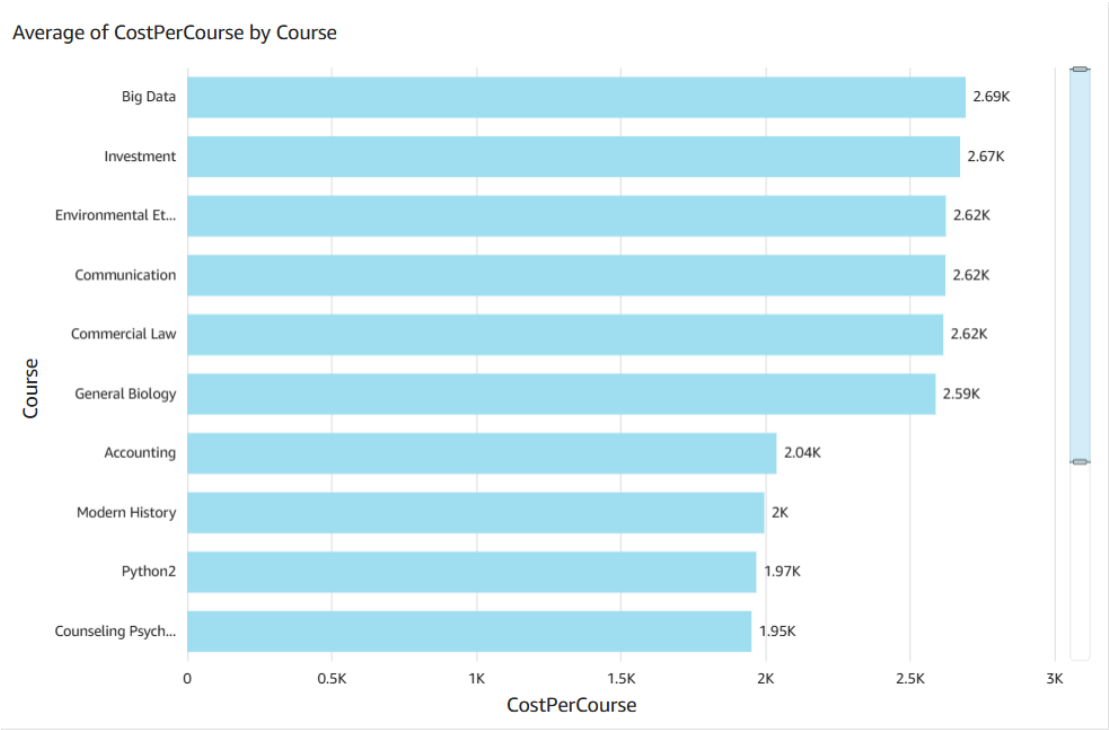
1



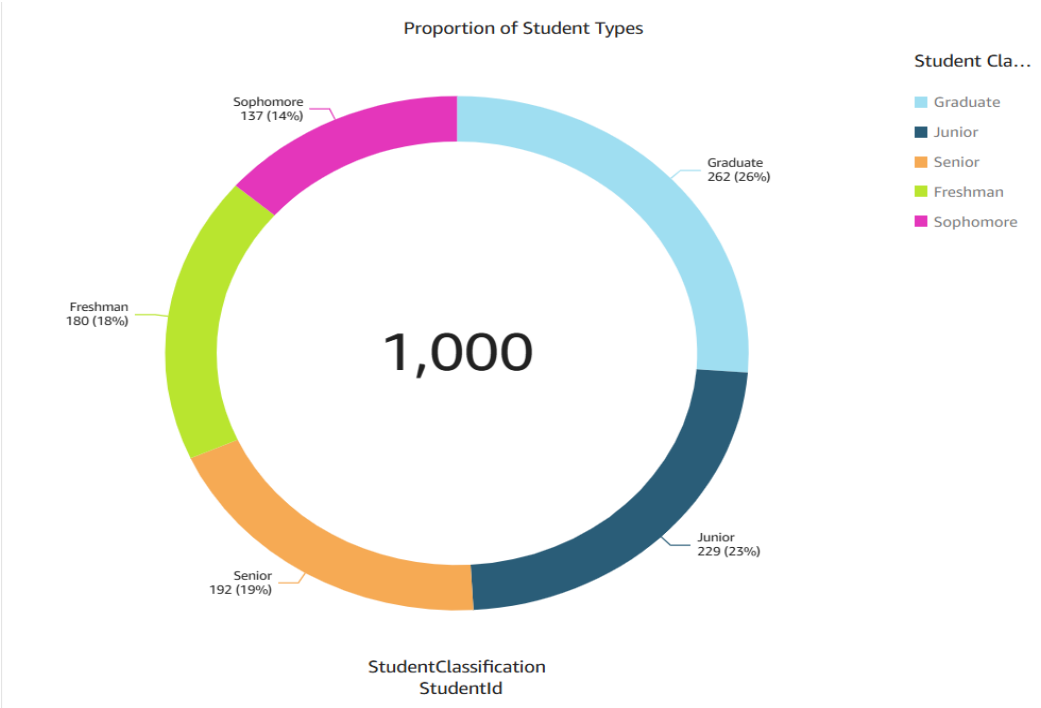
This bar chart presents the distribution of students across different majors, grouped by academic year. It allows the college to identify which majors are consistently popular, which ones are growing in demand, and which may be declining. By reviewing these enrollment trends over time, administrators can make better decisions about where to allocate resources, which programs may need more faculty, and which areas may require marketing or curriculum updates. This visual is crucial for long-term planning and aligning academic offerings with student interests.



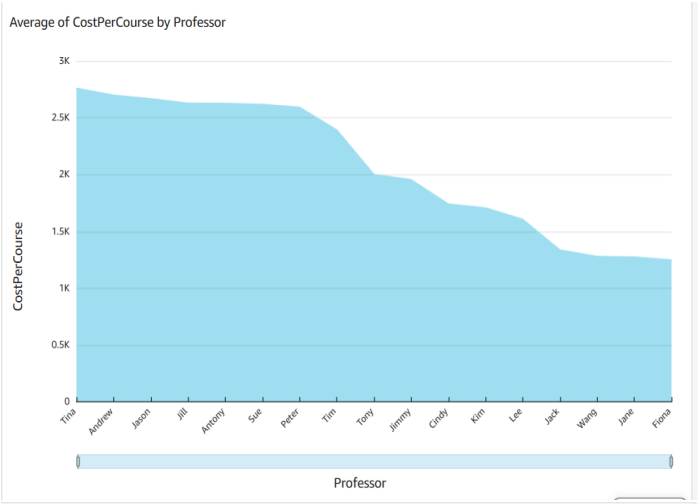
This chart shows the courses that received the highest average evaluation scores from students. High evaluation scores usually reflect student satisfaction with course content, teaching quality, and overall learning experience. This visual is valuable because it helps the college identify strong academic programs that can serve as models for others. It also highlights the courses where teaching practices, curriculum design, or student engagement strategies are most effective. By studying these top-performing courses, the institution can replicate successful practices across other departments and improve the overall quality of education.

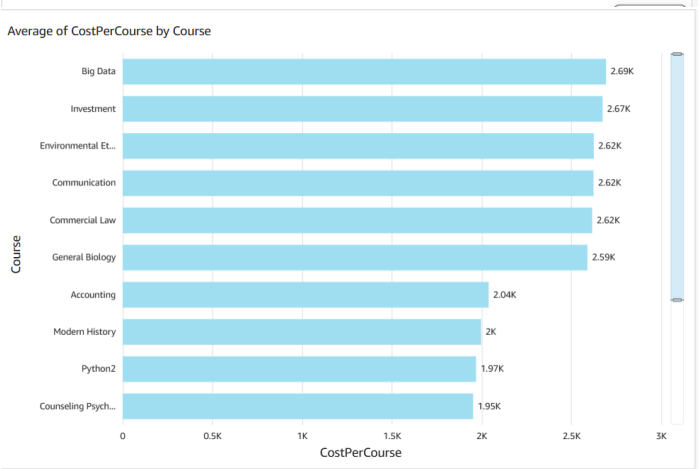


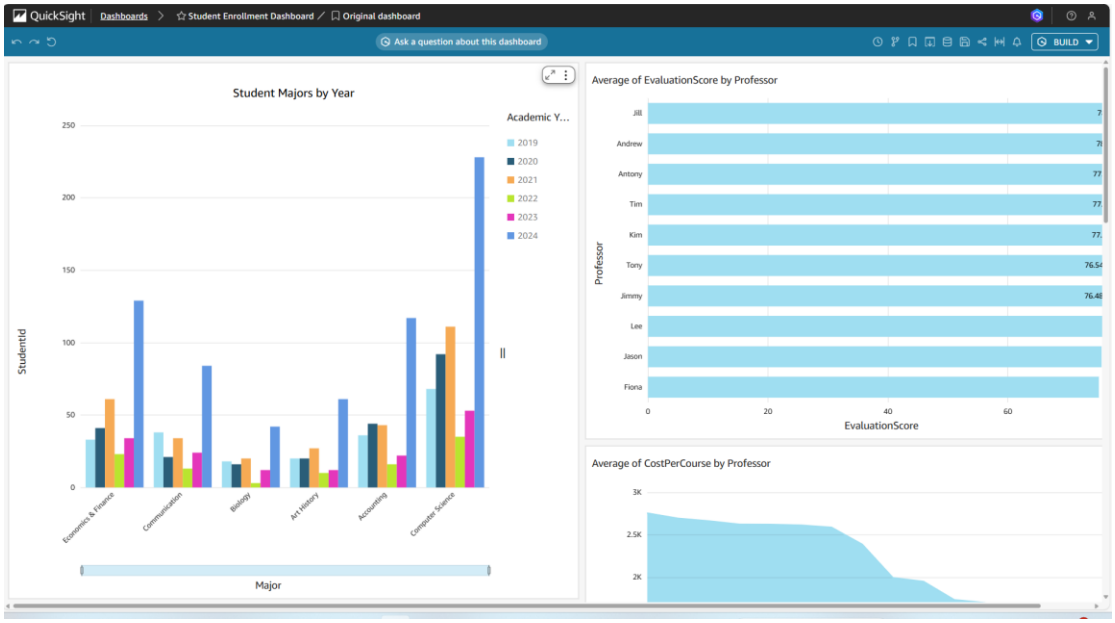
This visual identifies which courses are the most expensive on average. Understanding course costs is essential for balancing the college's budget and ensuring that tuition fees are set fairly. Expensive courses may require specialized equipment, lab resources, or highly qualified instructors, but they must also justify their value to students. By comparing cost data with evaluation scores and enrollment trends, decision-makers can evaluate whether high-cost courses are delivering strong outcomes. This visual supports financial planning, cost control, and ensures the sustainability of the college's academic programs.

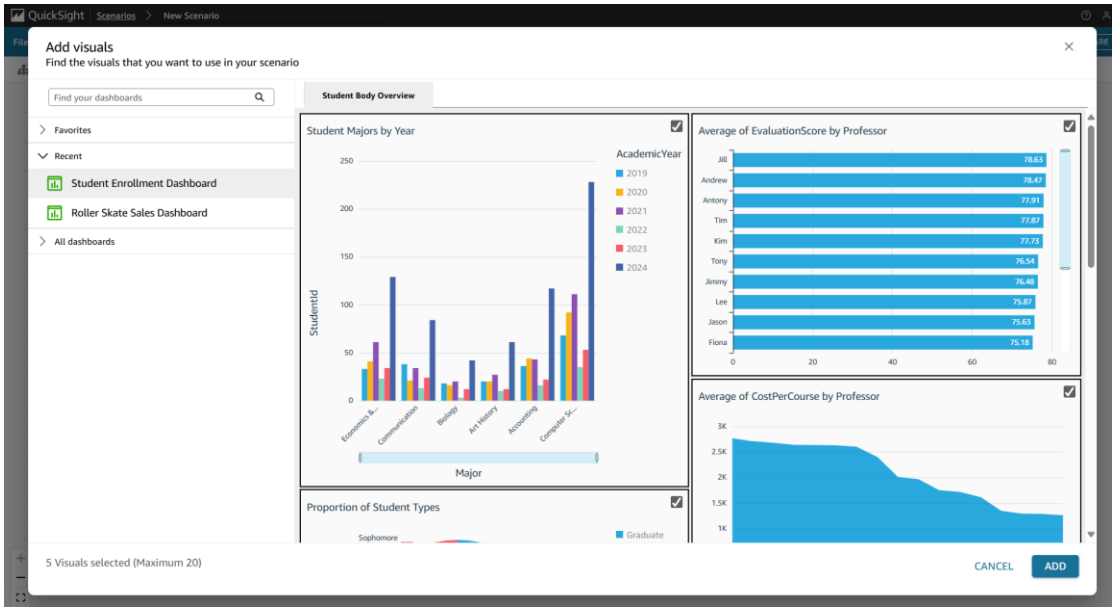


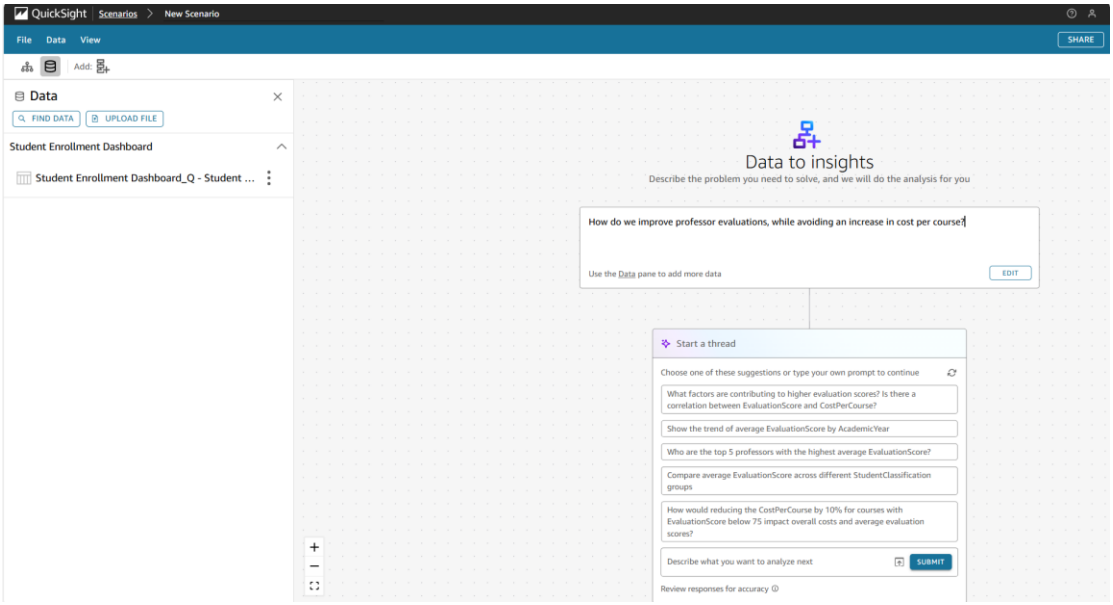
This pie chart illustrates the breakdown of students into two main categories: “Youth” (students under 30 years old) and “Adult Continuing Education” (students aged 30 and above). This information helps the college understand the composition of its student body. If a large percentage of adult learners is present, the institution may consider offering more flexible course schedules, evening classes, or online programs. On the other hand, a higher proportion of younger students may indicate a stronger need for student services, extracurricular programs, and academic advising. This chart provides insight into how age demographics influence the college’s strategic planning.

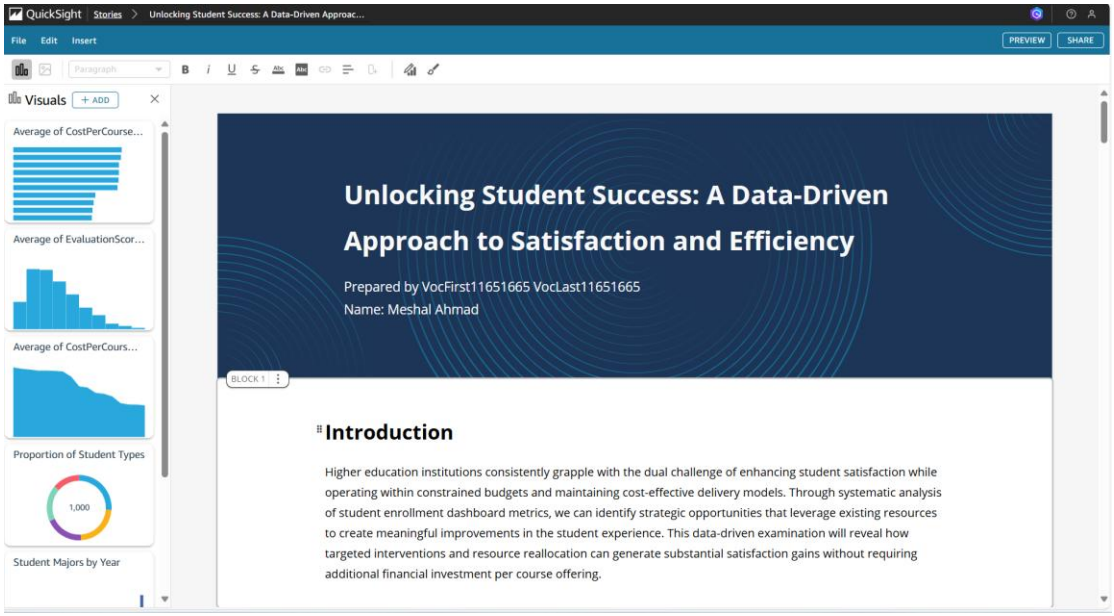

Average of CostPerCourse by Professor
CostPerCourse
3K
2.5K
2K
1.5K
1K
0.5K
0
Tina
Andrew
Jason
Jill
Antony
Sue
Peter
Tim
Tony
Jimmy
Cindy
Kim
Lee
Jack
Wang
Jane
Fiona
Professor


Average of CostPerCourse by Course
Course
Big Data
2.69K
Investment
2.67K
Environmental Et...
2.62K
Communication
2.62K
Commercial Law
2.62K
General Biology
2.59K
Accounting
2.04K
Modern History
2K
Python2
1.97K
Counseling Psych...
1.95K
0
0.5K
1K
1.5K
2K
2.5K
3K
CostPerCourse


QuickSight
Dashboards
Student Enrollment Dashboard
Original dashboard
Ask a question about this dashboard
BUILD
Student Majors by Year
Academic Y...
2019
2020
2021
2022
2023
2024
StudentId
250
200
150
100
50
0
Economics & Finance
Communication
Biology
Art History
Accounting
Computer Science
Major
Average of EvaluationScore by Professor
Jill
Andrew
Antony
Tim
Kim
Tony
Jimmy
Lee
Jason
Fiona
Professor
0
20
40
60
EvaluationScore
Average of CostPerCourse by Professor
3K
2.5K
2K


QuickSight
Scenarios
New Scenario
Add visuals
Find the visuals that you want to use in your scenario
Find your dashboards
Favorites
Recent
Student Enrollment Dashboard
Roller Skate Sales Dashboard
All dashboards
Student Body Overview
Student Majors by Year
AcademicYear
2019
2020
2021
2022
2023
2024
StudentId
Major
Average of EvaluationScore by Professor
Jill
Andrew
Antony
Tim
Kim
Tony
Jimmy
Lee
Jason
Fiona
78.63
78.47
77.91
77.87
77.73
76.54
76.48
75.87
75.65
75.18
Average of CostPerCourse by Professor
Proportion of Student Types
Sophomore
Graduate
5 Visuals selected (Maximum 20)
CANCEL
ADD


QuickSight
Scenarios
New Scenario
File
Data
View
SHARE
Add:
Data
FIND DATA
UPLOAD FILE
Student Enrollment Dashboard
Student Enrollment Dashboard_Q - Student ...
Data to insights
Describe the problem you need to solve, and we will do the analysis for you
How do we improve professor evaluations, while avoiding an increase in cost per course?
Use the Data pane to add more data
EDIT
Start a thread
Choose one of these suggestions or type your own prompt to continue
What factors are contributing to higher evaluation scores? Is there a correlation between EvaluationScore and CostPerCourse?
Show the trend of average EvaluationScore by AcademicYear
Who are the top 5 professors with the highest average EvaluationScore?
Compare average EvaluationScore across different StudentClassification groups
How would reducing the CostPerCourse by 10% for courses with EvaluationScore below 75 impact overall costs and average evaluation scores?
Describe what you want to analyze next
SUBMIT
Review responses for accuracy



"In this final scenario, all the visuals are displayed clearly and simply, as if presented to the Applied College."

Business Insights

- The college can focus resources on popular majors.
- High-cost courses can be reviewed for efficiency.
- Highly rated courses can be used as benchmarks.
- Age demographics help design flexible programs.

Conclusion

This project proves that AWS analytics tools can turn raw data into valuable insights that support academic and financial planning.